Day 57

* random module :-

→ random one of pre-defined module present in python.

→ The purpose of random is that "To generate random values in various contexts".

→ random module contains the following essential functions.

Integers
- a) randrange()
- b) randint()

floating
- c) random()
- d) Uniform()
- e) choice()
- f) shuffle()
- g) sample()

a) randrange()

⇒ This function is used for generating random integer values between specified limits.

syntax ①:- random. ~~range~~ randrange (value)

- This syntax generates any random value Between 0 to value-1.

syntax ②:- random. randrange (start, stop, step)

- This syntax generates any random value between start to stop-1

examples

```
>>> import random
>>> print (random.randrange (100,150)) - - - 133
>>> print (random.randrange (100,150)) - - - 121
>>> print (random.randrange (100,150)) - - - 106
>>> print (random.randrange (10)) - - - 5
>>> print (random.randrange (10)) - - - 9
```

#randrangeex.py

```
import random
for i in range (1,6):
        print (random.randrange (10))
print (" - - - - - - -")
for i in range (1,6):
        print (random.randrange (1000,1100))
print (" - - - - - -")
```

b) randint():-

Syntax: random.randint (start,stop)

- This syntax generates any random value Between start to stop. Here start and stop are inclusive.

examples)

```
>>> print (random.randint (10,15)) - - - 10
>>> print (random.randint (10,15)) - - - 13
>>> print (random.randint (10,15)) - - - 15
```

#randintex.py

```
import random
for i in range (1,6):
        print (random.randint (10,20))
print (" - - - - - ")
```

c) random():-

Syntax:- random.random()

- This syntax generates Floating point random values Between 0.0 and 1.0 (exclusive))

examples

```
>>> import random
>>> print (random.random()) - - - - 0.1623489 6503
>>> print (random.random()) - - - 0.153863298.
```

randomex.py

```
import randomm
lst = []
for i in range (1,6):
        lst.append ("%0.2f" % randomm.randomm ())
print ("- - - - - - -")
print ("Content of lst = {}".format (lst))
```

→ print (round (random (),2))

d) uniform ()

syntax : random.uniform (start, stop)

⇓

⇒ This generates random Floating point values from start to stop-1 values.

⇒ This value of start and stop can both integer or Floating point values.

examples

```
>>> import random
>>> print (random.uniform(10,15)) - - - 14.14638325.
>>> print (random.uniform (10,15)) - - - 13.186384.
>>> print (random.uniform (10.75, 15.75) - - - - 13.9386543289
>>> print (random.uniform (10.75, 15.75) - - - - 12.683432.
```

uniformex.py

```
import random
lst = []
for i in range (1,6):
        lst.append (float ("%0.2f" % random.uniform (10,15.5)))
print ("- - - - - - ")
print ("Content of lst = {}".format (lst)).
```

e) choice ():

syntax : random.choice (iterable_object)

⇓

• This functions obtains random values from iterable_object example

```
>> print(random.choice([10,20,30,40,50]), random.choice("PYTHON"),
                                          random.choice(range(10,15)))
                     -- 40 T 11
```

#choiceex.py

```
import random
s = "AaBRe#^%@8YuaLPav*&"
for i in range(1,6):
    print(random.choice(s), random.choice(s), random.choice(s),
                                          random.choice(s))
```

f) shuffle:-

→ This function is used for re-organizing the elements of any mutable object but not on immutable object.

syntax :- random.shuffle(list)

⇒ we can shuffle the data of list but not other objects of Data Types.

example

```
>>> d = {10:"cadbury", 20:"kitkat", 30:"malky bar", 40:"dairymilk"}.
>>> print(d) - - - {10:'cadbury', 20:kit ↑ same it
>>> for k,v in d.items():
        print(k, "--", v)

    → 10 -- cadbury.
      20 -- kitkat
      30 -- milky bar
      40 -- dairymilk.

>>> import module
>>> l = [10,20,30,40,50].
>>> print(random shuffle(l) --- None).
>>> print(l) ---- [30,40,50,10,20].
>>> random.shuffle(l) ---- [40,10,20,50]
```

#shuffleex).

```
import random as r
l = [10,"python","rossum", 34.56, True].
for i in range(1,6):
        r.shuffle(l)
    print("content of l =",l).
```

8). sample()

=> This Function is used for selecting random samples from any iterable object based on number of samples (+ve)

syntax: random.sample(iterable_object, k)

=> Here 'k' can be number of samples.

Examples

```
>>> import random
>>> s = "ABCabcBRTYvertyuf%^&*#@!%^&ghjkiyl"
>>> print(random.sample(s,5)) --- ['A','m','n','j','t']
```

sampleex.py

```
import random
lst = [10, "Rossum", "python", 34.56, True]
for i in range(1,6):
        print(random.sample(lst,2))
```

ex①. shuffle

```
import random as r
s = "HYDERABAD"
lst = list(s)
for i in range(1,len(s)):
        r.shuffle(lst)
        print(k)
```

sample

```
from random import sample
s = "PYTHON"
for i in range(1,6):
    xprint(sample(s,k=3))        k=""
    k = ""
    lst = samples(s, k=3)
    k = k.join(lst)
    print(k).
```

ex2.
```
import random as r
Cap = "ABCDEFGHIJKL...Z 0123456789 abcdefghijklmn..
!@#$%^&*(), +="

for i in range(1,10):
    lst = r.sample(Cap,k=5)
    k = ""
    k = k.join(lst)
    print(k)
```

ex3.
```
import random as r
state = ["TS", "AP"]
stcode = ["09","08","07","06","39"]
nums = "0123456789"
alphs = "ABCDEFGHL...Z"
# numberplate : TS08 HF 8558.
for i in range(1,10):
    st1 = r.sample(state,1)
    rc = r.sample(stcode,1)
    lets = r.sample(alphs,2)
    no = r.sample(nums,4)
    sst1 = ""
    sst1 = sst1.join(st1)
    src = ""
    src = src.join(rc)
    slets = ""
    slets = slets.join(lets)
    sno = ""
    sno = sno.join(no)
    numberplate = sst1+src+" "+slets+sno
    print(numberplate)
```

random module

a.randrange()

- syntax:- random.randrange(value)
- syntax:- random.randrange(start,stop,step)

In [2]:
```
import random
```

In [3]:
```
random.randrange(10)
```

Out[3]: 3

In [4]:
```
random.randrange(10,15)
```

Out[4]: 11

In [6]:
```
random.randrange(100,150)
```

Out[6]: 129

In [7]:
```
random.randrange(100,150)
```

Out[7]: 130

In [8]:
```
random.randrange(100,150)
```

Out[8]: 146

In [9]:
```
random.randrange(10)
```

Out[9]: 5

In [11]:
```
import random
for i in range(1,6):
    print(random.randrange(10))
print("------------------------------------")
for i in range(1,6):
    print(random.randrange(100,1001))
print("--------------------------------------")
```

```
1
2
7
0
5
------------------------------------
652
791
788
334
321
--------------------------------------
```

b.randint()

- syntax:- random.randint(start,stop)

In [12]:
```
import random
```

In [14]:
```
random.randint(10,100)
```

Out[14]: 73

In [15]:
```
random.randint(10,100)
```

Out[15]: 86

In [16]:
```
random.randint(10,100)
```

Out[16]: 82

In [18]:
```
import random
for i in range(1,6):
    print(random.randint(10,100))
print('--------------------------------')
```

```
28
68
44
82
41
--------------------------------
```

c.random()

- syntax:- random.random()

In [19]:
```
import random
```

In [20]:
```
random.random()
```

Out[20]: 0.3390893584997424

In [21]:
```
random.random()
```

Out[21]: 0.6940686718132116

In [22]:
```
import random
for i in range(1,6):
    print(random.random())
print("-------------------------------------")
```

```
0.9823950874728699
0.30573095091646485
0.15091895604770267
0.6663381589310605
0.24247293938704373
-------------------------------------
```

In [26]:
```
import random
lst=[]
for i in range (1,6):
    (lst.append("%0.2f"%random.random()))
print("-------------------------------")
print("content of lst={}".format(lst))
```

```
-------------------------------
content of lst=['0.18', '0.98', '0.48', '0.35', '0.19']
```

d.uniform()

- syntax:- random.uniform(start,stop)

In [27]:
```
import random
```

In [28]:
```
random.uniform(10,100)
```

Out[28]: 32.807642980363404

In [29]:
```
random.uniform(10,100)
```

Out[29]: 17.21937622727144

In [30]:
```
random.uniform(10,100)
```

Out[30]: 44.87155426507271

In [31]:
```
import random
lst=[]
for i in range(1,6):
    lst.append(float("%0.2f"%random.uniform(10,100)))
print("--------------------------------------------------------")
print("content of list={}".format(lst))
```

```
--------------------------------------------------------
content of list=[34.91, 90.26, 73.86, 90.62, 28.98]
```

e.choice()

- syntax:- random.choice()

In [33]:
```
print(random.choice([10,20,30,40]),random.choice(['apple','banana','grapes']),ra
```

```
30 banana 4
```

In [34]:
```
import random
s="Abhbfdbvjhhlkshliflnfkdnfndlk%@88789478_)(*&@$#^$%%^**)"
for i in range(1,6):
    print(random.choice(s))
print("----------------------------")
```

```
f
k
$
*
%
----------------------------
```

f.shuffle()

- syntax:- random.shuffle(list)

In [35]:
```
d={10:'cadburry',20:'kitkat',30:'perk',40:'dairy milk'}
print(d)
```

```
{10: 'cadburry', 20: 'kitkat', 30: 'perk', 40: 'dairy milk'}
```

In [36]:
```
for k,v in d.items():
    print(k,"------------>",v)
```

```
10 ------------> cadburry
20 ------------> kitkat
30 ------------> perk
40 ------------> dairy milk
```

In [38]:
```
import random
```

In [40]:
```
l=[10,20,30,40,50,60]
print(random.shuffle(l))
```

```
None
```

In [41]:
```
print(l)
```

```
[10, 30, 20, 50, 60, 40]
```

In [45]:
```
print(random.shuffle(l))
```

```
None
```

In [51]:
```
random.shuffle(l)
```

In [54]:
```
import random
l=[10,"python","rossum",34.56,5+3j,True]
for i in range(1,6):
    random.shuffle(l)
print("content of l=",l)
```

```
content of l= [34.56, (5+3j), 'rossum', True, 10, 'python']
```

g.sample()

- synatax:- random.sample(iterable_object,k)

In [55]:
```
import random
```

In [58]:
```
s="nsjdjjashdgnrlkehioharzsl@#$$#$$^%^%*&*)(4864351564456"
print(random.sample(s,5))
print(random.sample(s,5))
```

```
[')', '5', '$', '&', 'k']
['s', 'g', 'j', 'k', '@']
```

In [59]:
```
import random
lst=[10,"rossum","python",34.56,True]
for i in range(1,6):
    print(random.sample(lst,2))
```

```
[True, 'rossum']
['python', 10]
[True, 10]
[True, 34.56]
[True, 34.56]
```

In [ ]: